four exchange data centers;[2272] as well as $189,342 in costs that are common to all competing consolidators, as described above.

The Commission estimates that the total direct costs to each existing exclusive SIP that will decide to enter as a competing consolidator will include $5.63 million in ongoing annual costs, as discussed above, and total one-time costs of approximately $3 million. The one-time costs per existing exclusive SIP are composed of labor costs of $1,394,300;[2273] external costs of $1,237,500 to modify its systems to comply with Rules 614(d)(1) through (d)(4), external costs of $14,000 to purchase market data from the SROs, an additional initial external cost of $194,000 to co-locate itself at four exchange data centers;[2274] as well as $189,342 in costs that are common to all competing consolidators, as described above.

The Commission estimates that the total direct costs to each new entrant that is not an SRO or a data aggregator, in the competing consolidator space and to each SRO that will decide to enter as a competing consolidator will include approximately $5.63 million in ongoing annual costs, as discussed above, and total one-time costs of approximately $5.66 million.[2275] The one-time costs are composed of labor costs of $2,788,600,[2276] external costs of $2,475,000 to build its systems to comply with Rules 614(d)(1) through (d)(4), external costs of $14,000 to purchase

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2272 Id.

2273 See supra Section IV.D.3(d)(iii).

2274 Id.

2275 The Commission believes that competing consolidators that are affiliated with exchanges will choose to operate under the provisions of the exemption. See supra Section V.C.2(a)(i)a.

2276 See supra Section IV.D.3(e)(iii).

market data from the SROs, an additional initial external cost of $194,000 to co-locate itself at four exchange data centers;[2277] as well as $189,342 in costs that are common to all competing consolidators, as described above.

One commenter stated that the Proposing Release underestimates the direct costs to become a competing consolidator. The commenter said that "[t]he Commission estimates that potential competing consolidators would incur 'total one time costs of up to between approximately $897,000 and $2.40 MM, depending on entity type.' Even the higher end of that range is a fraction of what ICE believes it would cost to build the necessary infrastructure to be a competing consolidator."[2278] On the other hand, one commenter stated that the existing exclusive SIPs would have a competitive advantage over other potential competing consolidators, "because they would not incur the upfront capital expenditures to build a Competing Consolidator model."[2279] While acknowledging that some potential competing consolidators might incur lower costs than others to become a competing consolidator, the

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2277 Id.

2278 The commenter also stated that "the capital expenditure costs to build the NMS network were estimated at $3.8 million, and the ongoing costs to maintain and operate the NMS network are estimated to be $215,000 annually." See IDS Letter I at 13. The Commission believes this is informative but not directly applicable to the costs that potential competing consolidators could incur when building or modifying their systems to operate as a competing consolidator for two reasons. First, unlike the potential competing consolidators with one of their main functions being data consolidation, the NMS network is not a system that consolidates market data. Second, the NMS network costs include the transmission of options data, which competing consolidators will not consolidate or disseminate.

2279 See MIAX Letter at 2–3.

Commission is revising up its cost estimates from the Proposing Release.[2280] The Commission estimates that the direct costs to each potential competing consolidator will be between approximately $5.6 million in ongoing annual costs, and total one-time costs of up to between approximately $1.7 million and $5.7 million, depending on entity type.[2281]

(ii) Direct Costs to SROs

Separately, the Commission estimates that the SROs will jointly have approximately $175,000 in direct one-time costs and approximately $102,000 in ongoing costs for the amendments to the effective national market system plan(s).[2282] These costs include the costs SROs will incur when conducting an assessment of competing consolidator performance and developing an annual report of such assessment to be provided to the Commission. Additionally, each SRO will incur approximately $71,000 in one-time direct costs[2283] and approximately $128,000 in ongoing costs for the collection and dissemination of information necessary to generate consolidated market data required by Rule 603(b).[2284]

One commenter mentioned that the amendments will "require the SROs to continue to incur costs associated with managing an NMS plan while overseeing and reporting on competing consolidators."[2285] The requirement that the SROs conduct an assessment of and report on

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2280 See Proposing Release, 85 FR at 16843 for a discussion on the costs to becoming a competing consolidator.

2281 See supra note 2256 for a discussion on costs including the costs to comply with Regulation SCI.

2282 See supra Section IV.D.6(c).

2283 See supra Section IV.D.7(a)(iii); supra note 1553.

2284 See supra Section IV.D.7(b)(iii); supra note 1559.

2285 See NYSE Letter II at 28.

competing consolidators' performance is new and the Commission did not include ongoing direct costs from this requirement to the SROs in the Proposing Release. However, with the amendments, the Commission revises its estimates to include $102,165 of ongoing direct costs jointly incurred by the SROs.[2286] The SROs and the Operating Committee will have access to information made publicly available by competing consolidators, which could be used as part of their assessment of competing consolidators. The SROs can mitigate some of their costs by using this information.

The commenter also stated that the Commission "also places on exchanges the costs of calculating and disseminating certain regulatory data (such as LULD bands) to competing consolidators and self-aggregators."[2287] The commenter said that the cost to obtain data from other exchanges needed to perform these calculations should be considered by the Commission. The Commission disagrees that calculation of the regulatory data required by this rule will impose any major new data costs on the exchanges and the Commission's estimates of the costs to collect and disseminate this certain regulatory data are included in the estimates of direct costs to the SROs. The national securities exchanges currently aggregate market data obtained from the exclusive SIPs and from proprietary data feeds to perform several exchange functions, including order handling and execution, order routing, and regulatory compliance. Therefore, if they continue to use the same proprietary data for their regulatory data calculations, there would

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2286 This ongoing direct cost number is calculated using the PRA ongoing burden hours for maintaining the required timestamps, conducting assessments of competing consolidators, preparing an annual report, maintaining the list of the primary listing exchange for each NMS stock, and calculating gross revenues (Attorney at $417 for 245 hours equals $102,165). See supra Section IV.D.6(c).

2287 See NYSE Letter II at 20–21.

not be major new costs. To the extent they can use the new consolidated market data to perform the regulatory data calculations, the exchanges can become self-aggregators[2288] and benefit from potentially lower data content costs.

(iii)Indirect Costs to the Exclusive SIPs

The Commission believes that the amendments may impose a substantial cost for existing exclusive SIPs in terms of loss of data processing revenues because exclusive SIPs will no longer be exclusive consolidators and disseminators of market data, and at least one of the exclusive SIPs–Nasdaq UTP–will no longer be paid out of the plan for its processing costs.[2289] The Commission believes that the exclusive SIPs' loss of revenue will be mitigated by the opportunity to become competing consolidators. If the exclusive SIPs decide to become competing consolidators, they will compete for business with each other and with other competing consolidators. This competition may lead to revenue that is lower than their current revenues. This potential decrease in revenue will represent a transfer of resources to other competing consolidators and to market participants potentially increasing social welfare. On the other hand, if the exclusive SIPs decide to become competing consolidators, their experience with this market may give them a competitive advantage and help mitigate their potential revenue losses. The exclusive SIPs have the benefit of having been in this business for a long time. They have significant connectivity to market participants and vendors and can leverage their existing customer base and established relationships with vendors and purchasers at firms.

2288 See supra Section III.D.2(a) for a discussion on the scope of the definition of self-aggregator.

2289 This does not apply to CTA/CQ Plan that, as discussed above, is paid differently. See supra Section V.B.2(d).

Additionally, as mentioned below, the CTA SIP received some improvements from recent investments.

(iv) Direct and Indirect Costs to Certain Market Participants and Investors

The Commission believes that the amendments are likely to have indirect costs—such as potentially paying for unused data content and implementation costs of switching from SIP or proprietary data to consolidated market data—to certain market participants and investors.

First, the Commission believes that there will be an implementation cost for market participants to switch from using current exclusive SIP providers or proprietary data feeds to using competing consolidators. This cost is likely to vary among types of market participants; for instance, existing purchasers of proprietary DOB data products are likely to assume limited additional costs while new customers of consolidated market data products from competing consolidators will need, for example, to establish new connectivity and integrate a larger set of data into their operations. This implementation cost will include administrative costs for subscribing to a new provider of the data, as well as any infrastructure investments that may be needed to handle the data as delivered by the competing consolidator. One of the commenters stated that the cost to replace or integrate a new data feed might be approximately $1 million and that "[s]maller firms would try to do the same at lower cost."[2290] The Commission is uncertain whether the cost number mentioned in this comment letter covers costs to get this new feed from a competing consolidator or from the exchanges directly. The Commission believes that the ultimate size of these costs will likely vary by market participant. For example, for market participants that currently use proprietary data feeds and that will continue to use their existing systems and infrastructure after switching to consolidated market data, these costs are likely

[2290] See Data Boiler Letter I at 79–80.

lower. On the other hand, for market participants who need to build brand new systems and infrastructure to be able to receive consolidated market data, these costs could be higher and closer to the number the commenter states.

Additionally, one of the current exclusive SIPs, SIAC, processes and disseminates the academic TAQ dataset. If SIAC discontinues its SIP business, there may be interruptions to the availability of this data, which will create a cost for both the academic community and investors that otherwise benefit from academic and regulatory use of this dataset and the research derived from it. On the other hand, other data vendors also provide comprehensive historical data products and that may become more readily or more affordably available from competing consolidators, especially because competing consolidators do not have to take in all data content underlying consolidated market data and offer a data product with the entirety of consolidated market data.[2291] The Commission is uncertain and acknowledges the possibility that TAQ may no longer be available and consolidated market data products may not be affordable to the academic community.[2292] The Commission is unable to quantify the incremental social welfare cost of the interruption of availability of the TAQ dataset.

Some commenters stated that the decentralized consolidation model will increase costs for market participants because they would have to contract with a backup competing consolidator to avoid disruptions should their primary competing consolidator experience a

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2291 See, e.g., MayStreet, Market Data, available at http://maystreet.com/products/market-data/ (last accessed Jan. 2, 2020).

2292 One commenter stated that "NYSE's TAQ product is licensed to the academic community at a steep discount to its true cost." See Wharton Letter at 2.

disruption.[2293] One commenter said that "[i]n a world of multiple consolidators, business continuity concerns will force many market participants to subscribe to more than one consolidator as a backup."[2294] Other commenters stated that in absence of a backup, a competing consolidator's customers would be significantly impacted by a disruption of their original data source.[2295]

Under the amendments, market participants will not be required to incur the costs of retaining a back-up competing consolidator, though some may choose to do so after evaluating the needs of their business and their customers. Currently, many market participants that rely on proprietary data use SIP data as their back-up[2296] and market participants that rely on SIP data do not have a back-up option besides the exclusive SIPs' geographically diverse back-up system as required by Regulation SCI. Under amendments, market participants subscribing to "SCI competing consolidators" will similarly benefit from the requirements that those competing consolidators have geographically diverse backup and recovery capabilities, pursuant to Regulation SCI.

On the other hand, market participants that will receive consolidated market data products from competing consolidators that are not SCI entities might decide to maintain a connection to a back-up competing consolidator (i.e., from a secondary source) based on their business needs. The Commission is uncertain what costs may be associated with the need for

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2293 See, e.g., Angel Letter at 20; NYSE Letter II at 24; FINRA Letter at 4.

2294 See Angel Letter at 20.

2295 See NYSE Letter II at 24; Nasdaq Letter IV at 8, 36.

2296 One commenter said that some of the ATSs use SIP data "as a backstop" to their proprietary data feeds. See BestEx Research Letter at 3.

backup competing consolidators in the decentralized consolidation model, but does not believe they are necessarily higher than costs to maintain backups today. This is because the new competing consolidator business might generate a secondary market where some competing consolidators compete to provide backup options, which might lower backup costs.[2297] For example, some competing consolidators might provide a backup option with narrower data content and higher latency, similar to the current SIP data. But unlike the exclusive SIPs providing the current SIP data, these competing consolidators would be under competitive pressure and would be more likely to provide cheaper backup data and connectivity options than the SIP data. New backup costs will likely differ for different market participants and will be affected by the new competitive competing consolidator market as well as the new market data fees, both of which will have pricing decisions to make about the provision of backup services. The costs may also depend on decisions competing consolidators may make regarding the resiliency of their own products and what backup requirements would be necessary for their customers in light of such decisions.

(v) Indirect Costs to SROs

One commenter said that the Proposing Release "requires SROs to 'make available' to every competing consolidator and self-aggregator 'all data necessary to generate consolidated market data'—but does not make clear how SROs would be compensated for the cost of delivering such market data information."[2298] The Proposing Release discusses that the SROs

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2297 One commenter said that "[t]he existence of multiple SIP vendors will allow firms to choose the best offering for their purposes and others as backstops, reducing the reliance on a single SIP feed vendor." See BestEx Research Letter at 5.

2298 See NYSE Letter II at 21.

will receive data content fees and connectivity fees as well as how the various fees will need to be filed.[2299] As discussed above, the SROs are allowed to provide their core data to competing consolidators and self-aggregators via the existing proprietary data feeds, a combination of proprietary data feeds, or a newly developed core data feed.[2300] While the SROs are not required to, if they choose to offer core data via a newly developed core data feed, they might incur some development costs to provide that new data feed. However, the Commission believes that the SROs may not have an incentive to develop a dedicated core data feed because they could incur costs of doing so.[2301] For example, if an SRO developed a dedicated core data feed, the SRO would have to take steps to ensure that any proprietary data feed is not made available on a more timely basis (i.e., by any time increment that could be measured by the SRO) than a core data feed. This means that if the core data feed were slower than the proprietary data feeds, the exchange would need to throttle any order-by-order proprietary data feeds.[2302] An exchange lowering its proprietary data speeds might also increase the number of market participants that

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2299 The fees for consolidated market data content will be established by the effective national market system plan(s) and file with the Commission under Rule 608. Each SRO will have to file with the Commission any proposed new fees for connectivity to its individual data that underlies consolidated market data pursuant to Section 19(b) of the Exchange Act and Rule 19b-4 thereunder and any proposed connectivity fee must satisfy the statutory standards. See Proposing Release, 85 FR at 16769, n.433.

2300 See supra Section III.B.9. See also supra note 795 for a discussion on how competing consolidators and self-aggregators are permitted to choose among the data feed options offered by the SROs to provide consolidated market data.

2301 None of the commenters indicated that they would provide a dedicated core data feed instead of using their existing proprietary data infrastructure.

2302 See supra Section III.B.9(b).